Rubric

Total Points: 100 + 5 Extra Credit

TA Checks

Complete these checks before scoring.

Check 1: Readability

- Confirm that the submitted report PDF opens and is readable.
- Confirm that the submitted materials are sufficient to begin grading.

Action: If the report cannot be opened or evaluated, the assignment should receive a zero.

Check 2: Required Task

- Confirm that the report uses the Forest Cover Type / Covertype dataset.
- Confirm that the task is multiclass classification on `Cover_Type`.
- Confirm that the target classes are treated as classes 1 through 7, not collapsed into a binary target or treated as regression.

Action: If the report does not use Covertype or does not perform multiclass classification on `Cover_Type`, the assignment should receive a zero.

Check 3: Access

- Confirm that the report includes a read-only Overleaf link.
- Confirm that the Overleaf project uses the student's **Georgia Tech / Enterprise Overleaf account**, not a personal Overleaf account.
- Confirm that the reproducibility sheet includes a GitHub commit hash.
- Confirm that the repository uses the student's **GT Enterprise GitHub repository**, not a personal GitHub repository.
- Confirm that course staff can access the Overleaf project and GitHub repository without requesting edit permissions or personal-account invitations.

Action: If the required Overleaf project or GitHub repository is missing, inaccessible, or cannot be verified after escalation to the Head TA, the assignment may receive a zero for unverifiable provenance. If the artifacts are accessible but one access condition is incomplete or noncompliant, apply the scored Submission Access criteria and do not apply additional deductions.

Check 4: Page Limit

- Do not read or score content beyond the 8 page limit.

Action: If required content appears only after the page limit, treat that content as missing when applying the scored rubric.

Check 5: Draft Format

- Check whether the report reads like a formal technical paper rather than a checklist.
- Check for excessive bullet-only writing that replaces interpretation, especially in Results, Discussion, or Conclusion sections.
- Multiple instances of four or more consecutive bullets in analysis-heavy sections should be treated as excessive unless they are clearly part of a compact protocol, hyperparameter list, or reproducibility note.

Action: If the report contains excessive bullet-only writing that replaces interpretation, the report should not be scored and should be treated as a draft.

Rubric Points

For each of the following items, grade all-or-nothing. If an item is worth 2 points, award either 0 or 2 points. This helps maintain consistency across the grading staff.

Compliance (16 points)

Data Compliance (3 points)

- **(2 pts) Sample size.**
  Award if: The report uses an approximately 20,000-instance working dataset and documents the sample size clearly.
  Do not award if: The report uses the full dataset, an arbitrarily tiny subset, or a substantially different sample size without justification.
- **(1 pt) Source and target.**
  Award if: The report identifies the data source and uses `Cover_Type` classes 1 through 7 consistently.
  Do not award if: The data source is unclear or the label mapping is ambiguous.

Submission Access (4 points)

- **(2 pts) Enterprise Overleaf.**
  Award if: The report includes a working read-only Overleaf link associated with the student's Georgia Tech / Enterprise Overleaf account.
  Do not award if: The link is missing, broken, requires edit permissions, or appears to use a personal Overleaf account.
- **(2 pts) Enterprise GitHub.**
  Award if: The report or reproducibility sheet includes a valid commit hash from the student's GT Enterprise GitHub repository, and the repository is accessible to course staff.
  Do not award if: The hash is missing, invalid, inaccessible, or hosted only in a personal GitHub repository.

Reproducibility (3 points)

- **(2 pts) Run instructions.**
  Award if: The student provides clear commands, environment requirements, data paths, and seeds sufficient to reproduce outputs on a standard Linux machine.
  Do not award if: Instructions are missing or clearly insufficient to reproduce results.
- **(1 pt) Sampling procedure.**
  Award if: The student provides the sampling method, sample size, random seed, and enough detail or code to recreate the sampled dataset.
  Do not award if: Sampling is described vaguely or the sampled dataset cannot be recreated.

Figure Legibility (2 points)

- **(2 pts) Readable figures and tables.**
  Award if: Figures, tables, axis labels, legends, captions, and text are readable at normal grading zoom, and the visual evidence can be evaluated without excessive zooming or guesswork.
  Do not award if: Multiple figures or tables are too small, cropped, blurry, poorly labeled, or otherwise unreadable, making the visual evidence difficult to verify.

Attribution (4 points)

- **(2 pts) Outside sources.**
  Award if: The report includes at least two relevant peer-reviewed sources beyond course materials and uses them to justify choices or interpret results.
  Do not award if: Sources are missing, not peer-reviewed, irrelevant, only listed without use, or not connected to the report's methodology or interpretation. Textbooks, arXiv/preprint papers, blog posts, tutorials, documentation pages, lecture notes, and vendor materials may be useful background resources, but they do not satisfy the required peer-reviewed outside-source requirement.

- **(2 pts) AI usage statement.**
  Award if: The report includes a clear AI/LLM usage statement.
  Do not award if: The AI statement is missing, vague, does not identify tools used, does not describe how tools were used, or does not affirm that the student reviewed and understood all assisted content.

Study Design (14 points)

Goal: The report should justify the dataset sample, metrics, hypothesis, validation protocol, and leakage controls.

Extended EDA may appear in the reproducibility sheet, but the main report must include the evidence needed to justify sampling, metrics, hypotheses, preprocessing, and interpretation.

EDA Evidence (4 points)

- **(2 pts) Full-data EDA.**
  Award if: The report includes full-dataset shape, feature types, missingness or absence of missingness, class counts, and class proportions.
  Do not award if: Full-data EDA is absent, superficial, incorrect, or not connected to the task.

- **(2 pts) Sample EDA.**
  Award if: The report gives class counts and class proportions for the approximately 20,000-instance sample.
  Do not award if: Sample distribution is missing or unclear.

Sampling Design (4 points)

- **(3 pts) Sampling rationale.**
  Award if: Stratification is used and justified, or non-stratified sampling is defended with evidence that all classes are preserved and the distribution is reasonable.
  Do not award if: The report states that data was sampled randomly without evidence, rationale, or comparison to the full class distribution.

- **(1 pt) Ordering effects.**
  Award if: The student uses a randomized or stratified procedure, or provides a compelling reason for a different choice.
  Do not award if: The student appears to use `head(20000)` or the first rows without justification.

Metrics and Hypothesis (4 points)

- **(2 pts) Metric justification.**
  Award if: The report explains why Accuracy, Macro-F1, balanced accuracy, confusion matrix, and per-class precision/recall/F1 are appropriate and why accuracy alone is insufficient.
  Do not award if: Metrics are merely listed without explanation or only accuracy is meaningfully discussed.
- **(1 pt) Early hypothesis.**
  Award if: At least one hypothesis is stated after EDA and before tuned results, predicts model behavior or trade-offs, and is tied to dataset properties and course theory.
  Do not award if: The hypothesis is just a goal, is not testable, or is stated after results.
- **(1 pt) Hypothesis revisit.**
  Award if: The report evaluates the hypothesis using metrics, learning curves, model-complexity curves, runtime, or error patterns.
  Do not award if: The hypothesis is never revisited or is ignored in discussion.

Validation (2 points)

- **(1 pt) Test holdout.**
  Award if: The test set is held out and used only once for final reporting.
  Do not award if: Hyperparameters are chosen based on test performance or the final test set is repeatedly used for model selection.
- **(1 pt) Training-only validation and leakage controls.**
  Award if: Hyperparameter tuning uses cross-validation or a fixed validation split on training data only, and preprocessing such as scaling is fit only on training data or training folds.
  Do not award if: Validation is unclear, appears to include test data, or preprocessing appears to be fit before splitting.

Decision Trees (14 points)

- **(2 pts) Complexity control.**
  Award if: The report identifies a control such as `max_depth`, `min_samples_leaf`, or `ccp_alpha`, and explains how it limits overfitting.
  Do not award if: No pruning or regularization mechanism is used or explained.
- **(3 pts) Learning curve.**
  Award if: The curve shows training and validation metric versus at least four training sizes and is used to discuss bias/variance.
  Do not award if: The learning curve is missing, has fewer than four training sizes without justification, is mislabeled, or is not interpreted.
- **(3 pts) Complexity curve.**
  Award if: The curve varies at least three reasonable values of a pruning or regularization parameter and supports the final model choice.
  Do not award if: The curve is missing, uses too few values without justification, varies a parameter unrelated to complexity, or is not interpreted.
- **(2 pts) Final evaluation.**
  Award if: Accuracy, Macro-F1, and balanced accuracy are reported, and confusion matrix or per-class precision/recall/F1 is included for this model or in the cross-model multiclass evaluation.
  Do not award if: Required metrics are missing or error patterns are not discussed.
- **(2 pts) Runtime.**
  Award if: Fit and predict time are reported with at least one trade-off.
  Do not award if: Runtime is missing or only one timing value is reported without context.
- **(2 pts) Tuning quality.**
  Award if: At least three reasonable hyperparameter values are tested, and the final choice is justified using validation or model-complexity evidence.
  Do not award if: The final model appears arbitrary, only one setting is tested, or the tested values are too narrow to support the conclusion.

kNN (14 points)

- **(2 pts) Scaling and distance.**
  Award if: Scaling is leakage-safe, and the distance metric and weighting choices are stated.
  Do not award if: Scaling-sensitive preprocessing is missing, unclear, or fit before splitting.
- **(2 pts) k exploration.**
  Award if: At least three distinct k values are tested and include meaningfully different neighborhood sizes, such as small, medium, and large values.
  Do not award if: Only one k is tested, fewer than three values are tested without justification, or the tested values are too narrow to show bias/variance behavior.
- **(2 pts) Learning curve.**
  Award if: The curve shows training and validation metric versus at least four training sizes and is interpreted.
  Do not award if: The learning curve is missing, has fewer than four training sizes without justification, is mislabeled, or is not interpreted.
- **(2 pts) Complexity curve.**
  Award if: The curve shows validation metric versus at least three distinct k values and supports the final model choice.
  Do not award if: The curve is missing, uses too few k values without justification, or is not interpreted.
- **(3 pts) Final evaluation.**
  Award if: Accuracy, Macro-F1, and balanced accuracy are reported, and confusion matrix or per-class precision/recall/F1 is included for this model or in the cross-model multiclass evaluation.
  Do not award if: Required metrics are missing or error patterns are not discussed.
- **(2 pts) Runtime.**
  Award if: Fit and predict time are reported, and prediction-time cost is discussed.
  Do not award if: Runtime is missing or prediction-time cost is ignored.
- **(1 pt) Sensitivity discussion.**
  Award if: The report discusses sensitivity to irrelevant features, high dimensionality, mixed continuous/binary features, or class imbalance.
  Do not award if: No kNN-specific limitation is discussed.

SVM (14 points)

For computationally expensive SVM experiments, students may use a clearly justified validation strategy or reduced tuning grid, provided the comparison is reproducible and scientifically defensible.

- **(2 pts) Kernel comparison.**
  Award if: Results from at least two kernels, such as linear and RBF, are reported and compared.
  Do not award if: Only one kernel is used or the kernels are mentioned but not actually compared.
- **(2 pts) Scaling.**
  Award if: Scaling is leakage-safe and applied consistently.
  Do not award if: Scaling is missing, unclear, or fit before splitting.
- **(2 pts) Learning curve.**
  Award if: The curve shows training and validation metric versus at least four training sizes and is interpreted.
  Do not award if: The learning curve is missing, has fewer than four training sizes without justification, is mislabeled, or is not interpreted.
- **(2 pts) Complexity curve.**
  Award if: The curve varies C and/or γ across at least three reasonable settings total, or provides a justified smaller grid for computational reasons.
  Do not award if: The curve is missing, uses too few values without justification, varies an irrelevant parameter, or is not interpreted.

- **(3 pts) Final evaluation.**
  Award if: Accuracy, Macro-F1, and balanced accuracy are reported, and confusion matrix or per-class precision/recall/F1 is included for this model or in the cross-model multiclass evaluation.
  Do not award if: Required metrics are missing or error patterns are not discussed.

- **(1 pt) Runtime.**
  Award if: Fit and predict time are reported and kernel trade-offs are discussed.
  Do not award if: Runtime is missing or kernel runtime trade-offs are ignored.

- **(2 pts) Tuning quality.**
  Award if: At least two kernels are evaluated, C is varied for each kernel where applicable, and γ is varied for RBF or another nonlinear kernel where relevant. A smaller grid is acceptable only with runtime justification.
  Do not award if: The final model appears arbitrary, kernel choice is unsupported, or C/γ tuning is missing without justification.

Neural Networks (14 points)

If the student completes only one of the two required implementations, they should receive at most half credit for this section.

- **(3 pts) SGD setup.**
  Award if: Optimizer is explicitly SGD, no disallowed optimizers are used, and learning rate, batch size, epochs, early stopping if used, and regularization are documented for both implementations.
  Do not award if: Momentum, Nesterov, Adam, AdamW, Adagrad, RMSprop, or another adaptive optimizer is used, or the training setup is not documented.

- **(2 pts) Epoch curve.**
  Award if: The report shows train and validation loss or metric over epochs for both the scikit-learn and PyTorch neural-network implementations and interprets optimization behavior.
  Do not award if: Epoch evidence is missing, shown for only one implementation without justification, or not interpreted.

- **(2 pts) Learning curve.**
  Award if: The curve shows training and validation metric versus at least four training sizes and is interpreted.
  Do not award if: The learning curve is missing, has fewer than four training sizes without justification, is mislabeled, or is not interpreted.

- **(2 pts) Complexity curve.**
  Award if: The curve varies at least three reasonable neural network settings, such as width, depth, learning rate, or regularization, and supports the final model choice.
  Do not award if: The curve is missing, uses too few settings without justification, varies an irrelevant setting, or is not interpreted.

- **(2 pts) Final evaluation.**
  Award if: Accuracy, Macro-F1, and balanced accuracy are reported, and confusion matrix or per-class precision/recall/F1 is included for the neural network models or in the cross-model multiclass evaluation.
  Do not award if: Required metrics are missing or error patterns are not discussed.

- **(2 pts) Implementation comparison.**
  Award if: The report compares scikit-learn and PyTorch in terms of optimization behavior, stability, runtime, tunability, or implementation differences under SGD-only constraints.
  Do not award if: Both implementations are reported but not meaningfully compared.

- **(1 pt) Runtime and tuning.**
  Award if: At least three reasonable NN settings are tested, and fit/predict time or epoch-level training cost is reported and interpreted.
  Do not award if: The final model appears arbitrary, too few settings are tested without justification, or runtime/training cost is missing.

Discussion (14 points)

Goal: The report should move beyond presenting algorithm results and explain what the full set of experiments reveals about the dataset, the algorithms, and the modeling trade-offs.

The discussion should be written in paragraph form. It should not be a second results table or a bullet-only recap of scores.

- **(3 pts) Cross-model synthesis.**
  Award if: The discussion directly compares at least three of the required algorithm families and explains why their behavior differed using evidence from final metrics, learning curves, model-complexity curves, runtime, and/or error patterns. A strong discussion should connect observed behavior to course concepts such as bias–variance, locality, margin-based classification, scaling sensitivity, model capacity, or regularization.
  Do not award if: The report presents each model separately but does not compare them, only states which model scored highest, or gives unsupported claims such as "SVM performed best because it is powerful."
- **(3 pts) Performance and efficiency trade-offs.**
  Award if: The discussion explains trade-offs between predictive performance and computational cost. This should include at least two concrete comparisons, such as accuracy/Macro-F1 versus fit time, prediction time for kNN, SVM kernel cost, tree interpretability versus performance, or neural-network training stability versus runtime.
  Do not award if: The discussion reports runtime and metrics separately but does not explain trade-offs, ignores computational cost, or declares a winner using only one metric.
- **(3 pts) Class-level behavior.**
  Award if: The discussion analyzes multiclass performance rather than only aggregate scores. It should identify which cover types were easiest or hardest to classify, discuss minority-class behavior, and use the confusion matrix or per-class precision/recall/F1 to explain at least one important misclassification pattern.
  Do not award if: The discussion ignores class imbalance, only reports overall accuracy, does not mention minority classes, or includes a confusion matrix without interpreting class-level errors.
- **(3 pts) Curve-based interpretation.**
  Award if: The discussion uses learning curves and model-complexity curves to explain model behavior. It should make at least two specific claims about underfitting, overfitting, variance, bias, saturation with more data, or hyperparameter sensitivity, and those claims should be tied to the plotted curves.
  Do not award if: Curves are included but only described visually, such as "the line went up," or the discussion does not use the curves to support claims about bias, variance, underfitting, overfitting, or tuning behavior.
- **(2 pts) Limitations and improvements.**
  Award if: The discussion identifies at least one concrete limitation of the study and at least one evidence-based improvement or follow-up experiment. Acceptable limitations include the 20,000-instance sample, minority-class instability, limited SVM grid size due to runtime, SGD-only neural-network constraints, class imbalance, or restricted feature engineering. The improvement must follow from the evidence, such as testing class weighting after poor minority recall, expanding the SVM grid after sensitivity to C, increasing minority-class representation, or revisiting NN learning rates after unstable epoch curves.
  Do not award if: Limitations are missing, generic, or disconnected from the results. Improvements such as "use more data," "try more models," or "tune better" do not receive credit unless tied to specific evidence from the report.

Extra Credit: Activation Study

Up to 5 points. Extra credit must follow the SGD-only rule. If a disallowed optimizer is used, no extra credit should be awarded.

- **(2 pts) Activation curves.**
  Award if: Epoch-wise validation curves are shown for all four activations using the same training budget.
  Do not award if: Curves are missing, not comparable, or use different training protocols.

- **(2 pts) Comparison table.**
  Award if: A final comparison table reports metrics, runtime, and convergence notes.
  Do not award if: The comparison table is missing or incomplete.
- **(1 pt) Interpretation.**
  Award if: The report explains convergence stability and trade-offs under SGD.
  Do not award if: The activation study only reports numbers without interpretation.

Version Control

- v1.0 – 05/19/2026 – TJL refactored SL rubric for Summer 2026 Covertype assignment.

Rubric written by Theodore LaGrow.